

APURVE KARANWAL

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PROFESSIONAL SUMMARY

Forward-thinking software engineer focused on Full-Stack, Web3, and AI/ML architecture. Passionate about creating secure, highly scalable solutions that bridge decentralized technologies with intelligent systems and robust backend infrastructures.

TECHNICAL ARSENAL

Backend, Databases & Media Pipelines: Node.js, Express.js, FastAPI, Flask, MongoDB, Redis, Docker, Multer, Cloudinary.

Experienced in deploying containerized microservices, high-speed caching networks, and optimized multimedia upload pipelines.

Full-Stack & Frontend: React, Next.js, TypeScript, Tailwind CSS, Vite.

Architected complex split-pane WYSIWYG editors and highly responsive, real-time micro-frontend dashboards.

AI/ML & Computer Vision: PyTorch, TensorFlow.js (MoveNet), MediaPipe, LangGraph, OpenCV.

Proven capability in building real-time skeletal tracking, NLP diet parsers, and LLM-orchestrated autonomous agents.

Web3 & Applied Cryptography: Solidity, Ethereum, Ethers.js, SHA-256, LSB Steganography.

Demonstrated expertise in decentralized ownership registries and adversarial AI defense systems.

Core Languages: Python, JavaScript/TypeScript, C, Solidity.

KEY PROJECTS

PulseAI | Full-Stack Health Platform

[Next.js](#), [Node.js](#), [Python](#), [LangGraph](#), [MoveNet](#), [MediaPipe](#)

- Architected a microservices-based AI health platform combining real-time computer vision, NLP diet tracking, and LLM-powered coaching via a Flask/Django API gateway.
- Engineered a 5-layer computer vision accuracy engine utilizing TensorFlow.js and MoveNet to track fine-grained physiological markers (eyes and hands) for real-time posture correction and form scoring.
- Deployed a context-aware health coaching agent using LangChain and LangGraph for persistent session memory, integrated with an NLP-driven nutritional analysis pipeline.

MemePage | Decentralized Application (In Progress)

[Node.js](#), [Express](#), [React](#), [MongoDB](#), [Redis](#), [Docker](#), [Multer](#), [Cloudinary](#), [Ethereum](#)

- Architecting a full-stack, time-locked crypto investment platform leveraging Ethereum to allow users to stake capital securely on user-generated media.
- Engineering a high-performance backend utilizing Express and Node.js, featuring Redis for high-speed vote caching, MongoDB for scalable data storage, and Docker for containerization.
- Integrating Multer and Cloudinary for seamless, optimized image uploading and media management within the React frontend ecosystem.

PNGProtect | Enterprise Image Security

[FastAPI](#), [PyTorch](#), [OpenCV](#), [PyJWT](#), [Web3](#)

- Engineered a high-performance REST API utilizing adversarial AI (FGSM perturbations) to create an "AI Shield," preventing unauthorized machine learning model training on digital assets.
- Implemented invisible LSB steganography and SHA-256 cryptographic hashing for tamper-proof watermarking, ensuring robust persistence against compression and format conversion.
- Integrated a Web3 ownership registry using Ethereum smart contracts, enabling decentralized, auditable proof of ownership and secure metadata management.

mdfmt (README Studio) | Open-Source Tool

[React](#), [TypeScript](#), [Vite](#), [TipTap](#), [Tailwind CSS](#), [Zustand](#)

- Developed an open-source, split-pane WYSIWYG Markdown editor enabling instant, real-time rich-text to GitHub-Flavored Markdown conversion via Turndown.
- Engineered a client-side export pipeline utilizing the Blob API for secure, zero-latency .md file generation without server round-trips.

EDUCATION & ACADEMIC EXCELLENCE

KIET Group of Institutions B.Tech in Information Technology	2025 – 2029
Little Angels School • Academic Distinction: Graduated with top-tier marks, securing 95.5% in Class X and 83.8% in Class XII. • Early Engineering Foundation: Pioneered self-taught learning in software and hardware prior to university, developing foundational Python skills alongside hands-on embedded electronics projects.	2009 – 2024

EXPERIENCE & LEADERSHIP

Ambassador / HeLa Labs • Evangelize decentralized technologies and actively drive engagement within the HeLa Web3 ecosystem, expanding the developer community by curating technical learning resources.	2026 – Present
Delegate for Brazil / UNHRC Simulation • Debated and drafted policy proposals regarding global human rights and digital surveillance, demonstrating strong public speaking, research, and negotiation skills.	Mar 2026